Sprint 1 Work Breakdown Structure

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Overview

This document breaks Sprint 1 (User Authentication) into tasks using a failing-tests-first approach. The Backend and Frontend Test Matrix PDFs are written first, with every case set to Failing. Each deliverable is then built in the order Supabase migrations, backend, frontend, followed by integration tests for the whole deliverable. A final end-to-end task tests Sprint 1 as a whole. When all tests pass, the Status column in both matrices is updated to Passing.

Only Sprint 1 is covered here. Other deliverables and sprints are not repeated.

Output Location

All Sprint 1 documents are placed in:

```
documents/sprints/sprint-1/
  Sprint1_Work_Breakdown_Structure.pdf
  Sprint1_PERT_Timeline.pdf
  Sprint1_Backend_Test_Matrix.pdf
  Sprint1_Frontend_Test_Matrix.pdf
```

Test Matrix Format

Each matrix is a table with the fields below. Test IDs use the layer and deliverable, for example BE-REG-01 (backend, Registration) or FE-LOG-03 (frontend, Login & Session).

ID	Test Case	Expected Result	Status
BE-REG-01	Short description of the scenario	Observable outcome that defines pass	Failing

Status values: Failing (default, written before code), Passing, Blocked.

Sprint 1: User Authentication (parent task)

1. Sprint 1 Documents

- **1.1** Sprint 1 WBS (PDF)
- **1.2** Sprint 1 PERT Timeline (PDF)
- **1.3** Backend Test Matrix (PDF) - covers Registration, Login & Session, Password Recovery, Profile Management

- **1.4** Frontend Test Matrix (PDF) - covers Registration, Login & Session, Password Recovery, Profile Management

2. Registration

2.1 Supabase Migrations

- **2.1.1** Users Table Migration
- **2.1.2** Projects Table Migration (minimal, for Inbox) - see Notes

2.2 Backend

- **2.2.1** Account Creation API (Password Hashing, Input Validation)
- **2.2.2** Username & Email Availability Check
- **2.2.3** Default Inbox Project Creation

2.3 Frontend

- **2.3.1** Sign-Up UI

2.4 Integration Tests

- **2.4.1** Registration Backend Integration Tests (API + Supabase)
- **2.4.2** Registration Frontend Integration Tests (Sign-Up UI + API)

3. Login & Session

3.1 Supabase Migrations

- **3.1.1** No new migration required - uses Users Table from 2.1.1

3.2 Backend

- **3.2.1** Login API (Credential Verification, Input Validation)
- **3.2.2** Session Token Handling (Token Utilities)

3.3 Frontend

- **3.3.1** Login UI
- **3.3.2** Session Restore
- **3.3.3** Logout

3.4 Integration Tests

- **3.4.1** Login & Session Backend Integration Tests (API + Supabase)
- **3.4.2** Login & Session Frontend Integration Tests (Login, Session Restore, Logout + API)

4. Password Recovery

4.1 Supabase Migrations

- **4.1.1** Password Reset Table Migration

4.2 Backend

- **4.2.1** Forgot Password API (Input Validation)
- **4.2.2** Reset Link Email
- **4.2.3** Reset Token Validation API
- **4.2.4** Reset Password API (Password Hashing, Input Validation)

4.3 Frontend

- **4.3.1** Forgot Password UI
- **4.3.2** Reset Password UI

4.4 Integration Tests

- **4.4.1** Password Recovery Backend Integration Tests (API + Supabase + Email)
- **4.4.2** Password Recovery Frontend Integration Tests (Forgot/Reset UI + API)

5. Profile Management

5.1 Supabase Migrations

- **5.1.1** No new migration required - uses Users Table from 2.1.1

5.2 Backend

- **5.2.1** Update Profile API (Input Validation)

5.3 Frontend

- **5.3.1** Profile Settings UI

5.4 Integration Tests

- **5.4.1** Profile Management Backend Integration Tests (API + Supabase)
- **5.4.2** Profile Management Frontend Integration Tests (Profile Settings UI + API)

6. Sprint 1 System Tests

- **6.1** Sprint 1 End-to-End Tests (Registration, Login & Session, Password Recovery, Profile Management in one browser flow)

Notes

- Default Inbox Project Creation (2.2.3) needs a projects table. Task 2.1.2 adds a minimal one so Sprint 1 can stand alone; the full Project Data Model remains in Sprint 2. Remove 2.1.2 if the projects table is built earlier.
- Session Restore (3.3.2) and Logout (3.3.3) are frontend tasks that rely on the token handling built in 3.2.2.
- The original Authentication Testing group (Backend Auth Test Suite, Frontend Auth Test Suite, Auth Test Matrix) is replaced by the test matrix PDFs (1.3, 1.4), the per-deliverable integration tests, and the end-to-end task (6.1).